warning was published in Braginsky (1967).

372 [However, in 1976,. . . *uncertainty principle*.] The clarified warnings were published in Braginsky (1977) and Giffard (1976), and the uncertainty principle origin of the limit was explained in Thorne, Drever, Caves, Zimmermann, and Sandberg (1978).

375 [Roughly 10^{-21} was the answer,] See, e.g., the quasi-transcript of a 1978 conference discussion in Epstein and Clark (1979).

375 [We both found the answer . . . different routes.] Braginsky, Vorontsov, and Khalili (1978); Thorne, Drever, Caves, Zimmermann, and Sandberg (1978).

378 [In principle it would be possible to widen the bars' bandwidths] Michelson and Taber (1984).

383 [Interferometers for gravitational-wave detection . . . as did Robert Forward and colleagues] Gertsenshtein and Pustovoit (1962), Weber (1964), Weiss (1972), Moss, Miller, and Forward (1971).

383 [and Drever had added . . . to their design.] See, e.g., Drever (1991) and references therein.

387 [he redirected most of his own team's efforts . . . and modest funds.] See Braginsky and Khalili (1992).

391 [A key to success in our endeavor . . . or *LIGO*.] For an overview of the plans for LIGO see Abramovici et al. (1992).

CHAPTER 11

397 General comment about Chapter 11: The (rather minor) historical aspects of this chapter are based on (i) my own personal experience as a participant, (ii) my interviews with two other participants (INT-Damour, INT-Wald), (iii) my reading of scientific papers the participants wrote, and (iv) my experience as a student in a course on paradigms and scientific revolutions taught by Thomas Kuhn at Princeton University in 1965.

401 [*The Structure of Scientific Revolutions*] Kuhn (1962).

403 [This freedom carries power.] Richard Feynman, one of the greatest physicists of our century, described beautifully the power of having several different paradigms at one's fingertips in his lovely little book *The Character of Physical Law* (Feynman, 1965). Note, however, that he never uses the word "paradigm," and I suspect that he never read Thomas Kuhn's writings. Kuhn described how people like Feynman operate; Feynman just operated that way.

403 [That is why physicists . . . supplement to it.] The flat spacetime paradigm was devised more or less independently by a number of different people; it is known, technically, as a "field theory in flat spacetime formulation of general relativity." For an overview of its history and concepts, see the following passages in MTW: Sections 7.1 and 18.1; Boxes 7.1, 17.2, and 18.1; Exercise 7.3. For an elegant generalization of it, which elucidates its relationship to the curved spacetime paradigm, see Grishchuk, Petrov, and Popova (1984).

406 [In 1971 Hanni and Ruffini, . . . Jeff Cohen] Cohen and Wald (1971), Hanni and Ruffini (1973).

407 [Five years later Roger Blandford . . . power jets] Blandford and Znajek (1977).

409 [During 1977 and 1978, Znajek and . . . pictorial interpretation:] Znajek (1978), Damour (1978).